



BCA 1st Semester

COLLEGE ASSIGNMENT

Name : _____
Roll No. : _____
Class : BCA 1st Semester
Subject : System Analysis & Design



SHOP – PRODUCT SALE & BILLING

Feature Set I



System Features



1 Customer Details

- Customer ID
- Customer Name
- Contact Number



2 Product Details

- Product ID
- Product Name
- Price

123

3 Quantity

- Enter and validate the quantity purchased



4 Sale

- Record the product sale and update the transaction



5 Bill Calculation

- Calculate Total Amount, applicable discount/tax if required, and Final Bill Amount

*Good
Systems Build
Better Business*



ALGORITHM (Step by Step)

1

START

Initialize the system and display the main menu (Generate Bill / New Sale).

2

Input Customer Details

INPUT

Enter Customer ID, Customer Name, Contact Number.

PROCESSING

Store customer details in the transaction.

3

Input Product Details

INPUT

Enter Product ID, Product Name, Price.

PROCESSING

Fetch product details from the product list and display the price.

4

Input Quantity

INPUT

Enter quantity purchased.

DECISION

Is the quantity valid? (> 0 and within stock)

NO

Show error message and ask again.

YES

Proceed to next step.

5

Record Sale

PROCESSING

Calculate Sale Amount (Price $\times$ Quantity).
Update product stock and save transaction in the database.

6

Calculate Discount (if applicable)

INPUT

Check if discount is applicable (e.g., special offer, loyalty card).

PROCESSING

Calculate Discount Amount.

7

Calculate Tax (if applicable)

INPUT

Check if tax is applicable (e.g., GST).

PROCESSING

Calculate Tax Amount.

8

Calculate Total Amount

CALCULATION

Total Amount = (Sale Amount – Discount) + Tax

9

Calculate Final Bill Amount

CALCULATION

Final Bill Amount = Total Amount (or apply other rules if needed)

10

Generate Bill

PROCESSING

Display bill with Customer Details, Product Details, Quantity, Discount, Tax, Total Amount and Final Bill Amount.

11

Update Transaction

PROCESSING

Save the completed sale record in the system and update stock.

12

Output

OUTPUT

Show success message: "Sale Completed" with bill details.

13

Ask for Next Transaction

DECISION

Do you want to make another sale?

NO

Go to Step 14 (Stop).

YES

Go to Step 2 (New Sale).

14

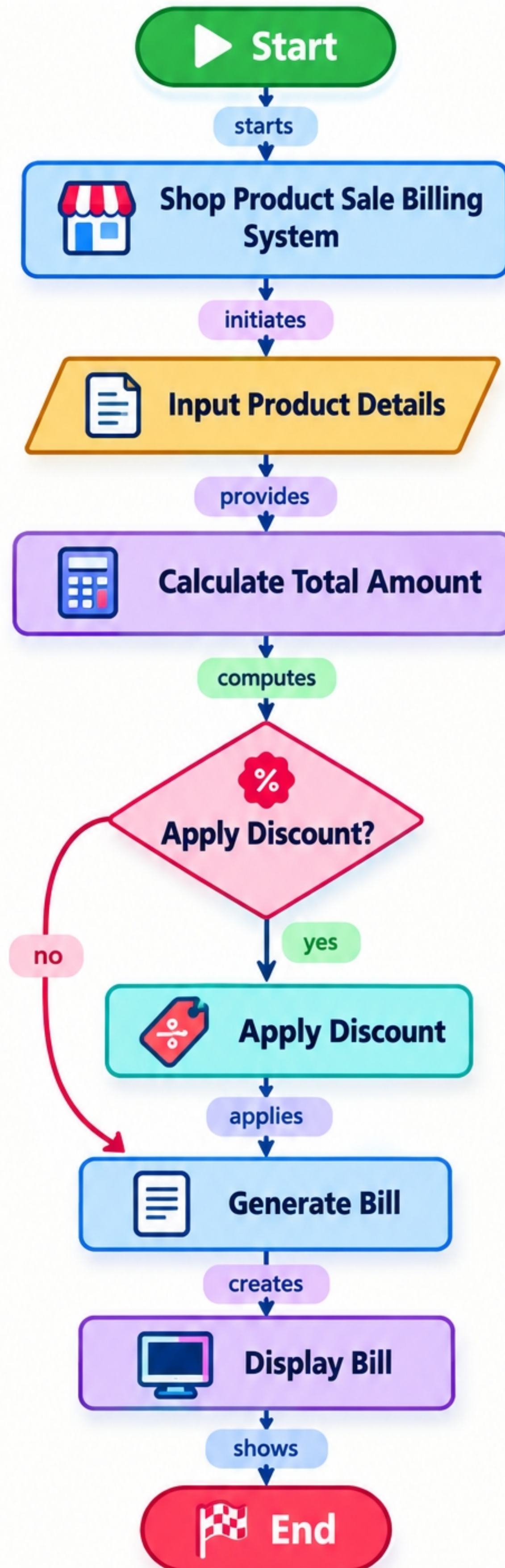
STOP



End the program.



Shop Product Sale Billing System





Shop Product Sale Billing System

CUSTOMER		
PK	int	Customer_ID
	string	Customer_Name
	string	Contact_Number

PRODUCT		
PK	int	Product_ID
	string	Product_Name
	decimal	Price

Legend

PK	= Primary Key
FK	= Foreign Key
int	= Integer
string	= String / Varchar
decimal	= Decimal
date	= Date



SALE		
PK	int	Sale_ID
	date	Sale_Date
	int	Customer_ID
	int	Product_ID
	int	Quantity
	decimal	Item_Amount

sold

in